Mohammadreza Hami

Computing Science MSc. Graduate

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Highlights of Skills

7 years of experience in AI and machhine learning research, including both theoretical and applied projects completed through advanced degrees and research assistantship.

- 7 years of AI and ML research and development experience polished through the completion of Bachelor's and Master's thesis, and employment as a Graduate Research Assistant.

Demonstrated hands-on proficiency in designing, testing, and fine-tuning prompts for LLMs

- Strong hands-on experience in prompt engineering and LLMs.

advanced capabilities in deep learning frameworks such as

- Advanced technical skills using Pytorch, Tensorflow, and Keras.
- Experienced in critical thinking, team work, and communication.

can we be just a bit more specific? Ability to lead projects, collaborate effectively in teams, and communicate complex ideas clearly.

Education

M.Sc. Computer Science (GPA: 4.0/4.0) *2022-2024*

University of Alberta, Edmonton, Canada.

add a space between educational pursuits

Thesis: Suboptimal Search with Dynamic Distribution of Suboptimality.

B.Sc. Computer Engineering (GPA: 18.14/20) *2018-2022*

University of Isfahan, Isfahan, Iran.

Thesis: Hashtag Generation for Twitter via Neural Topic Modeling.

Professional Experiences

Graduate Research Assistant, you don't need a comma here *01/2024 - present*

University of Alberta, Edmonton, Canada

Developed an innovative heuristic search algorithm that optimized performance metrics, achieving:

- Developed a new algorithm in Heuristic Search.

900% improvement over baseline algorithms in grid map scenari

enhancing efficiency in .search operations?

- 900% better performance than baselines on grid map domains.
- 400% better performance than baselines on exponential domains, demonstrating versatility across applications?

Add another bullet to corroborate your claim of team work skills, example: Collaborated with interdisciplinary teams to refine methodologies, contributing insight for publications and academic presentations.

Graduate Teaching Assistant, *09/2022 - 12/2024*

University of Alberta, Edmonton, Canada

- Introduction to Artificial Intelligence (Uninformed and Heuristic Search, Probabilistic Modeling and Reasoning, Bayesian Learning, Deep Learning, Reinforcement Learning and Multi-agent Systems)

Be specific about what you did: Assisted in instructing, provided one-on-on support, guided hands-on coding practice, as applicable.

- CMPUT 175 (3 times)

Data Engineer Internship *06/2021 - 08/2021*

Dataak, Tehran, Iran

since you're mentioning a language other than English, you could expand on this by stating your purpose (was it to adapt language models to handle language-specific nuances for example? To ensure accuracy in semantic understanding?

- Researched and developed various methods for Neural Topic Modeling in Persian Language
- Used Transformers, BERT, Topic modeling, Neural Networks, Large datasets of Tweets

Instructor of National Mathematical Olympiad Classes *09/2019 - 01/2021*

Iranian NODET high-schools

Maybe say you structured these classes by tailoring them to the rigorous standards of the National M. Oly.?

- Taught Combinatorics, Geometry and Algebra to students for the final round of National Mathematical Olympiad.

Were there any significant results you'd like to mention?

Publications

Suboptimal Search with Dynamic Distribution of Suboptimality, *12/2024*

Submitted to The 39th Annual AAAI Conference on Artificial Intelligence

Notable Projects

Revisiting Suboptimal Search [Link] *Winter 2023*
Single Agent Search course project, University of Alberta

- 50% better performance than Baselines.

Empirical Study of Differential Q-learning on Continuing Control Tasks [Link] *Fall 2022*
Reinforcement Learning course project, University of Alberta

Level Blending with Evolutionary Algorithm in Latent Space of VQ-VAE [Link] *Fall 2022*
Procedural Content Generation via Machine Learning course project, University of Alberta

Neural Hashtag Generator for Twitter using Transformers [Link] *Summer 2022*
Undergraduate Thesis project, University of Isfahan

- Used Sentence Transformers, BERTopic, Hugging Face, SKlearn

Winning multi-agent games with AI [Link] *Fall 2020*
Artificial Intelligence course project, University of Isfahan

- Used Q-Learning, A*, Genetic, and Mini-Max algorithms.

Honors and Awards

Graduate Research Assistantship Fellowship, *05/2023 - present*
Amii, Edmonton, Canada

Awarded Department Recruitment Scholarship, *09/2022*
University of Alberta, Edmonton, Canada

Ranked Top 4% **among all 96 Computer Engineering students,** *07/2022*
University of Isfahan, Isfahan, Iran

Awarded a membership in Exceptional Talents Organization, *07/2022*
University of Isfahan, Isfahan, Iran

Ranked 1st **in Iranian Competition Battle of Applications,** *10/2019*
University of Isfahan, Isfahan, Iran

Ranked Top 0.9% **in National Entrance Exam for B.Sc. Studies among all senior students,** *08/2018*
Tehran, Iran

Ranked 2nd **in International Mathematical Competition ToT,** *08/2018*
30th Summer Conference International Mathematical Tournament of Towns, Adygea, Russia

Passed the first round of National Informatics Olympiads, *2016 - 2017*
Tehran, Iran

Volunteering Experiences

Computer Science Instructor, *2019 - 2020*
Sharif University of Technology, Tehran, Iran

Head of ICPC Student Chapter, *2020 - 2021*
University of Isfahan, Isfahan, Iran

Skills

- **Languages:** Python, C++, C, JAVA, Matlab
- **Machine Learning:** PyTorch, Tensorflow, Keras, OpenAI Gym, Scikit-learn, NLTK
- **Tools & Frameworks:** LLMs, Transformers, Huggingface, RL-Glue, NumPy, Pandas, FastAPI, Scipy
- **Experienced in:** git, Jupyter, Bash, UNIX/Linux, Microsoft Windows, LaTeX